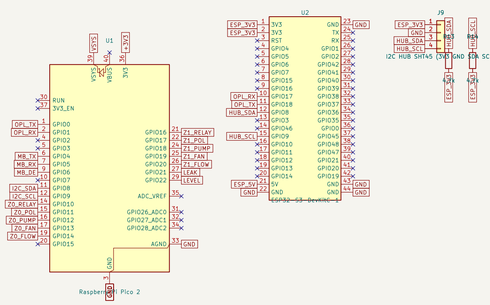
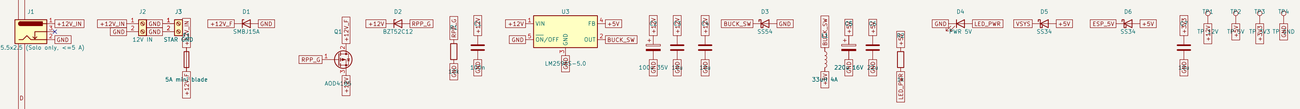
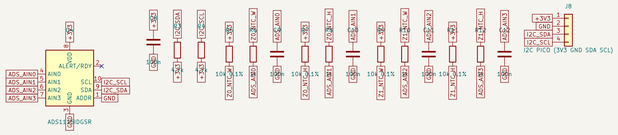
## Modules: Raspberry Pi Pico 2 (control MCU) + ESP32-S3-DevKitC-1 (hub), OPL UART GP0/GP1 <-> GPIO18/GPIO17



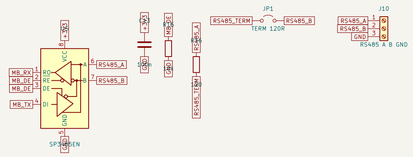

## 12 V input, fuse, TVS, reverse-polarity P-FET, 12->5 V buck (LM2596S-5.0), 5 V ORing, test points



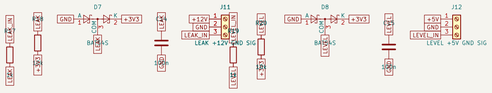

## ADS1115 @0x48 (ADDR=GND) on I2C0 GP8/GP9, NTC dividers 3.3 V — 10k 0.1 % — node — NTC — GND



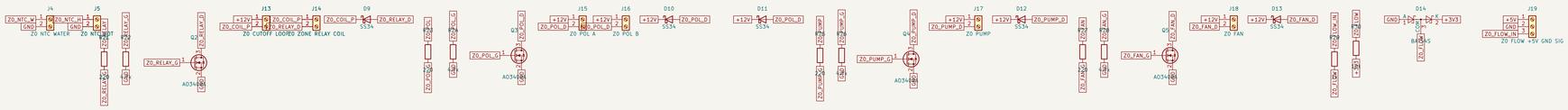

## RS-485 to the DPS modules: SP3485 on UART1 GP4/GP5, DE+/RE on GP6



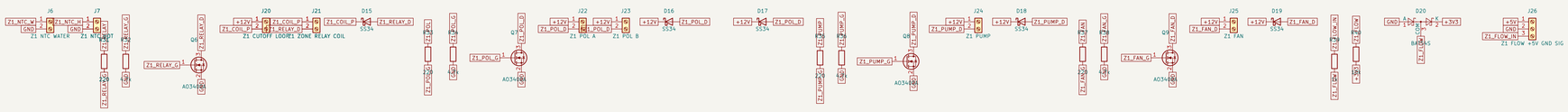

## Fail-safe inputs: LEAK GP21, LEVEL GP22 (LOW = good, open wire = HIGH = fault)




## Zone 0 drivers (GP10 relay, GP11 polarity, GP12 pump, GP13 fan, GP14 flow) and sensors




## Zone 1 drivers (GP16 relay, GP17 polarity, GP18 pump, GP19 fan, GP20 flow) and sensors




## Mounting holes




## Power flags




Pin map: firmware/opod-pico2/src/board.rs (source of truth)
Generated by hardware/pcb/carrier/scripts/gen_sch.py from design.py — do not edit by hand
Open Bed Climate — CERN-OHL-S-2.0
Sheet: /
File: carrier.kicad_sch

| | |
|---|---|
| **Title: Open Bed Climate — Tier 0 carrier, Rev A (UNTESTED)** | |
| Size: User | Date: |
| KiCad E.D.A. 10.0.0 | Rev: A<br>Id: 1/1 |